

MUHWEZI MARK

Telecommunications Engineer • Network Infrastructure & Backend Systems

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PROFILE

Telecommunications engineer who builds ISP infrastructure and the software that runs it. Currently provisioning FTTH subscribers at Savanna Fibre, with three years of field experience installing and troubleshooting Cambium microwave links, alongside a live multi-tenant billing and authentication platform serving hotspot operators in Uganda and Nigeria. Comfortable from RF alignment and RouterOS configuration through to Rust services, PostgreSQL and mobile money APIs. Degree completed at Kyambogo University, awaiting final results.

CORE SKILLS

Networking Cambium ePMP and PTP 550 microwave links, cnMaestro management, MikroTik RouterOS (hotspot, PPPoE, PPP authentication), FreeRADIUS, WireGuard, IP addressing & subnetting, CGNAT mitigation, link optimisation and fault diagnosis

Backend & Data Rust, Node.js, PostgreSQL, REST API design, Docker, Ubuntu server administration

Web & Mobile React.js, HTML/CSS, native Android (Java, Kotlin), Flutter cross-platform development

Cloud, Payments & APIs Firebase (Auth, Firestore, Cloud Storage, Hosting, Cloud Messaging), AWS, DNS and domain configuration, mobile money via pawaPay (MTN MoMo and Airtel Money), Africa's Talking (SMS, USSD), Resend transactional email, push notification delivery

Security Network threat modelling with Security Onion, Nmap and WireGuard

Field & Design Tools TEMS Investigation, walk test and drive test, Cisco Packet Tracer, AutoCAD, SolidWorks, EdrawMax

Embedded ESP32, Raspberry Pi, STM32

EXPERIENCE

Founder & Lead Engineer | Blossom Tech Solutions / TerraConnect Uganda, Kampala 2023 – Present

- Built and operate TerraConnect Uganda, a multi-tenant B2B platform giving ISP and WiFi hotspot operators billing, subscriber management, captive portals, voucher management and mobile money collections and payouts.
- Architected the backend in Rust and Node.js on PostgreSQL, deployed as Docker services on a production Ubuntu server, with FreeRADIUS integrated to MikroTik hotspot routers over WireGuard tunnels for centralised authentication and accounting.
- Integrated pawaPay for MTN MoMo and Airtel Money transactions, and completed a multi-round payment provider due diligence process covering AML/CFT policy, operator agreements and audit trails.
- Deployed and support seven live hotspot businesses across Kampala, handling everything from site survey and router provisioning to operator training and billing support.
- Delivered a router management and billing system plus a Flutter cross-platform application for QuickNet Networks, Nigeria (quicknetwifi.com).
- Published a native Android music player to Google Play and built client web platforms hosted on AWS.

FTTH Provisioning Engineer | Savanna Fibre Limited, Kampala 2026 – Present

- Provisioning fibre-to-the-home subscribers for a live Kampala ISP: installation, ONT configuration, activation and fault resolution at the customer premises.

PMP Field Engineer | Japotech Solutions Uganda Limited — contracted to Liquid Intelligent Technologies 2023 – Present

- Delivered field engineering on Liquid Intelligent Technologies' network as a Japotech contractor, joining as an intern and being recruited into the role.
- Installed 30 client links on Cambium ePMP and PTP 550 point-to-point and point-to-multipoint systems during 2025, covering site survey, mounting, alignment and commissioning, backed by Cambium vendor training in installation, link optimisation and cnMaestro management.
- Resolved 19 microwave link fault escalations, diagnosing interference, alignment drift, power and configuration issues to restore customer service.
- Carried out telecom site installations, walk tests and drive tests, and produced network survey documentation for the projects team.
- Awarded the company Certificate of Excellence for Integrity, 2024.

Intern — Field Maintenance & Transmission | Innovis Telecom Services International Ltd May – Jul 2025

- Completed a six-week structured rotation across field maintenance, transmission, data centre maintenance, network optimisation and drive test, and health and safety compliance, based at the Airtel Uganda MSC in Bugolobi.

EDUCATION

BSc Telecommunications Engineering | Kyambogo University 2022 – 2026, *awaiting results*

Final year project: TerraConnect Uganda — an automated router management system addressing the limitations of CGNAT. Supervised by Dr. Mugerwa Dickson.

Uganda Advanced Certificate of Education, Standard College Ntungamo, 2020 — principal passes in Mathematics, Physics and Chemistry.

Uganda Certificate of Education, Kitante Hill School, 2018 — Division 1, ten subjects passed.

CERTIFICATIONS & TRAINING

Cambium Networks Learning — vendor training completed 2024

- ePMP track: Installation Training for Force 180, 190, 200 and 300-25; Access Point Device Configuration; Link Optimization; Force 300-25 solution training
- PTP 550 track: Solution Overview; Configuring PTP 550; Configuring Antenna Gain; Understanding Metrics; Remote Management with cnMaestro

Other

- Internship Programme Completion Certificate — Innovis Telecom Services International Ltd, 2025
- Certificate of Excellence, Company Value of Integrity — Japotech Solutions, 2024
- Telecom site installation, point-to-multipoint installation, walk test and drive test — Japotech Solutions
- Website development, hosting and native Android development — Blossom Tech Solutions

OPEN SOURCE

1,134 contributions over the past year across 55+ repositories, including work on network backends, MikroTik integration tooling and trading systems — github.com/muhwezim78

LANGUAGES

English, Luganda and Runyankore — fluent in all three.

REFEREES

Mr. Jamil Wamala — Projects Team Leader, Japotech Solutions — +256 787 157 267

Mrs. Lyn Kyimezire — Assistant Projects Team Leader, Japotech Solutions — +256 757 052 015

Mr. Nyero Kenneth — Senior Developer, Blossom Tech Solutions — +256 758 830 464

Mr. Lutswala Emma — Field Engineer, Innovis Telecom Services International — +256 700 554 928